

Astrid Arena Olave Herrera

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EDUCATION

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| <p>Ph.D. in Mathematics
 <i>Michigan State University</i>
 Advisor: Elizabeth Munch</p> | <p>2021 - Present</p> |
| <p>M.Sc. in Applied Mathematics
 <i>Universidad Nacional de Colombia</i>
 Meritorious thesis : Revealing brain network dynamics during the emotional state of suspense using topological data analysis
 Advisor: José Perea (MSU). Coadvisor: Francisco Gómez</p> | <p>2021</p> |
| <p>B.Sc. with Honors in Mathematics
 <i>Universidad Nacional de Colombia</i>
 Thesis: Persistent homology analysis of RNA secondary structures
 Advisor: Gustavo N. Rubiano</p> | <p>2018</p> |

RESEARCH INTERESTS

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| <ul style="list-style-type: none"> • Topological data analysis:
Foundations and applications | <ul style="list-style-type: none"> • Algebraic topology • Mathematical biology |
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PAPERS

4. Astrid A. Olave. A new family of distances over partially ordered sets. *ArXiv e-prints*, June 2026, 2606.06377
3. Astrid A. Olave and Elizabeth Munch. Bounding the interleaving distance on concrete categories using a loss function. *ArXiv e-prints*, January 2026, 2601.09034
2. Astrid A. Olave, Jose A. Perea, and Francisco Gómez. Revealing brain network dynamics during the emotional state of suspense using tda. *Network Neuroscience*, page 1–25, August 2025. doi:10.1162/netn.a.34

1. Konstantinos Georgiou, Andy (Jia) Jiang, Edward Lee, Astrid A. Olave, Ian Seong, and Twesh Upadhyaya. Lift & project systems performing on the partial-vertex-cover polytope. *Theoretical Computer Science*, 820:1–16, June 2020. doi:10.1016/j.tcs.2020.03.004

PRESENTATIONS

11. *The topology of suspense: Using TDA to reveal brain network dynamics*. Talk in the Student Applied Mathematics seminar. Michigan State University. Sep 2025
10. *Bounding the interleaving distance on concrete categories using a loss function*. Talk in the Applied Algebraic Topology Research Network (AATRN). Online Seminar. Aug 2025
9. *Bounding the interleaving distance on concrete categories using a loss function*. Poster present at the workshop: The geometric realization of AATRN. Chicago, IL, US. Aug 2025
8. *Bounding the interleaving distance on concrete categories using a loss function*. Talk in the SIAM Conference on Applied Algebraic Geometry (AG25). Madison, WI, US. Jul 2025
7. *The history of the interleaving distance: A journey from homology to its applications*. Talk in colloquium. Universidad de Málaga, España. Jun 2025
6. *Generalizing the interleaving distance using categorical flows*. Short video on the AATRN tuthorial-a-thon Spring 2025. YouTube video. Feb 2025
5. *The history of the interleaving distance: A journey from homology in $\mathbb{R}^3$ to categorical flows*. Talk in the Student Geometry and Topology seminar. Michigan State University. Oct 2024
4. *Revealing brain network dynamics during the emotional state of suspense using topological data analysis*. Oral contribution presented at GEOTOP-A International Conference: Applications of Geometry and Topology. Mérida, México. Jan 2024
3. *Finding interleaving distances on a poset using categories with flows*. Poster present at 2023 AWM Research Symposium. Atlanta, GA, US. Oct 2023
2. *Revealing brain network dynamics during the emotional state of suspense using topological data analysis*. Short talk presented at II SIPAIM virtual Workshop. Online Seminar. May 2021
1. *Persistent homology analysis of RNA secondary structures*. Poster session presented at XXII Colombian Congress of Mathematics. Popayán, Colombia. Jun 2019

HONORS AND AWARDS

GTA Award for Excellence in Teaching <i>Michigan State University</i>	2025
Dr. Paul and Wilma Dressel Endowed Scholarship <i>Michigan State University</i>	2025
Mathematics Recruiting Fellowship <i>Michigan State University</i>	2021
Scholarship for complete tuition for M.Sc programs <i>Universidad Nacional de Colombia</i>	2019
Exemption from payment for best average <i>Universidad Nacional de Colombia</i>	2013-2018

TEACHING EXPERIENCE

Michigan State University

Matrix Algebra with Computational Applications (MTH 314) <i>Teaching assistant of section 004, Instructor of section 101</i>	Fall 2024, Summer 2023
Introduction to proofs (MTH 299) <i>Teaching assistant of sections 001 and 003</i>	Spring 2026, Fall 2025
Calculus III (MTH 234) <i>Teaching assistant of sections 001,015 and 018</i>	Spring 2024
Calculus II (MTH 133) <i>Teaching assistant of sections 013,018 and 024</i>	Spring 2023
Calculus I (MTH 132) <i>Teaching assistant of sections 019,021 and 024</i>	Fall 2022
Survey of Calculus I (MTH 124) <i>Teaching assistant of Lecture 1</i>	Fall 2021

College Algebra II (MTH 103B)

Spring 2025, Fall 2024

*Instructor of section 010, 001***Quantitative literacy (MTH 101)**

Summer 2026

*Teaching assistant of section 102***TEACHING ADJACENT EXPERIENCE**

Orientation Leader for First-Year PhD Students

2025 - 2026

Michigan State University

Organized monthly meetings to help incoming PhD students transition into the teaching and academic environment of MSU.

Leader for teaching assistants in MTH 234

Spring 2024

Michigan State University

Organized biweekly meetings to support teaching assistants and discuss lesson preparation

Conduct classroom observations

Fall 2023

Michigan State University

Provided teaching feedback for 8 teaching assistants

Mathematics Learning Center

Spring 2022

Michigan State University

Tutor

Integral calculus

First semester 2018

Universidad Nacional de Colombia

Writer of solutions of the book "Cálculo integral en una variable"

Mentorship**Summer Topology Program**

Summer 2026

Michigan State University

Guide undergraduate students study TDA.

Tutor of Círculos Matemáticos

Second semester 2019

Universidad Nacional de Colombia

Guided enthusiastic high school students about mathematics into the experience of being a mathematician

PROFESSIONAL CERTIFICATES

Certification in College Teaching

2024 - 2025

Michigan State University

Project: The implementation Khanmigo as a mathematics tutor:
A study from students' perspectives

Data Science Boot Camp

May - June 2024

The Erdős Institute

Project: Temporal Graphs for Music recommendation systems

SERVICE

Departmental service

AMS Graduate Student Chapter at MSU

2024-2025

Treasurer

Council of Graduate Students

2023-2024

*Representative of the mathematics department***Professional service**

Topological Data Analysis Seminar

2025-2026

Organizer

Graduate Student Top. and Geo. Conference

April 2024

*Organizer***Community outreach**

Marble Elementary School Math Night

November 2024

*Volunteer***Reviewer/Referee**

Symposium on Computational Geometry (SoCG)

2023

*Reviewer***EXTRACURRICULAR ACTIVITIES**

Intramural soccer

2021-2025

Player and captain of the mathematics team

Sin Medida inter-university catholic community

2018-2021

Leadership team member

Universidad Nacional women's soccer team

2014-2021

Goalkeeper